Connecting Neural Circuit Biotypes to Pharmacogenomics: An OWL 2 DL Ontology Extension for Precision Psychiatry Decision Support

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Brief Description of Work

Depression and anxiety disorders exhibit substantial biological heterogeneity, yet, to our knowledge, no existing system provides a unified, computable representation linking neural circuit biotype classification to pharmacogenomics safety constraints for antidepressant prescribing. We present a confidence-qualified neural phenotype ontology that connects these two domains through an OWL 2 DL knowledge architecture, a Bayesian network for biotype inference, and a pharmacogenomics clinical decision support module. This is an early-stage research system built to demonstrate that the semantic architecture patterns required for precision psychiatry decision support are tractable with current ontology engineering tools and open standards.

The core ontology formalizes relationships among neural circuit features (activation, task-free connectivity, task-evoked connectivity), dimensional psychopathology constructs (anhedonia, threat dysregulation, rumination, cognitive dyscontrol) drawn from the brain circuit taxonomy of Williams (2016) and Tozzi et al. (2024), treatment considerations, and evidence provenance, including reliability metadata from published test-retest studies (Elliott 2020). Following ontology-reuse best practice, it aligns to BFO 2020 upper-level categories and bridges via SKOS to UBERON (brain regions), HPO (phenotypes), the Cognitive Atlas (tasks), and SNOMED CT, reusing established identifiers where they exist and minting new URIs only for constructs that lack formal representations (e.g., RDoC domains, HiTOP spectra); SSSOM mapping tables accompany the release. A pgmpy Bayesian network computes posterior biotype membership probabilities from observed neural and behavioral features, with reliability flags propagated from the ontology. The system is validated as OWL 2 DL-conformant with zero ROBOT errors.

A pharmacogenomics extension module operationalizes the biotype-to-prescribing connection: an HL7 CDS Hooks 2.0 service integrates SPARQL-queried drug–gene relationships (OWL), CPIC Activity Score rules (Python rule engine; Relling 2011, Hicks 2015), and FHIR Genomics Reporting IG v3.0.0 genotype data to generate real-time antidepressant prescribing alerts that incorporate both biotype context and metabolizer phenotype. Prior pharmacogenomic CDS Hooks services exist (Dolin 2018); our contribution is layering probabilistic biotype context onto this deterministic safety pathway. A key architectural principle is the strict separation of ontology vocabulary (what exists), curated knowledge base (what is known), and procedural rule engine (what to do). This avoids the well-known anti-pattern of encoding IF/THEN clinical logic as OWL axioms, which would compromise DL decidability and make guideline updates brittle.

Link to Demo: https://sapphire-cision.github.io/bmds210-neural-phenotype-ontology/#brain

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